

Выполним эквивалентное преобразование схемы и определим комплексные параметры цепи

$$\omega = 2\pi f = 2\pi \cdot 500 = 3141.5927 \text{ rad/s}$$

$$E_1 = \frac{E_{m1}}{\sqrt{2}} e^{j\psi} = 21.213 e^{j45^\circ} \text{ V}$$

$$I_2 = \frac{I_{m2}}{\sqrt{2}} e^{j\psi} = 14.142 e^{-j45^\circ} \text{ A}$$

$$E_2 = \frac{I_2}{C_{m2}} = \frac{10 - j10}{0.2} = 50 - j50 \text{ V}, R_{m2} = \frac{1}{C_{m2}} = \frac{1}{0.2} = 5 \Omega$$

$$E_{eq} = E_1 - E_2 = 15 + j15 - (50 - j50) = -35 + j65 \text{ V}$$

$$Z_{eq, \text{н}} = (R_{m1} + R_{m2}) + jX_{Lm1} = (5 + 5) + j3 = 10 + j3 \Omega = 10.44 e^{j16.7^\circ} \text{ deg}$$

Рассчитаем эквивалентные сопротивления ветвей приемника

$$X_{L1} = \omega L_1 = 3141.5927 \cdot 0.0024 = 7.54 \Omega, X_{C1} = \frac{1}{\omega C_1} = \frac{1}{3141.5927 \cdot 0.000035} = 9.095 \Omega$$

$$X_{L2} = \omega L_2 = 3141.5927 \cdot 0.0015 = 4.712 \Omega, X_{C2} = \frac{1}{\omega C_2} = \frac{1}{3141.5927 \cdot 0.000045} = 7.074 \Omega$$

$$X_{L3} = \omega L_3 = 3141.5927 \cdot 0 = 0 \Omega, X_{C3} = \frac{1}{\omega C_3} = \frac{1}{3141.5927 \cdot 0.000038} = 8.377 \Omega$$

$$Z_1 = R_1 + jX_{L1} - jX_{C1} = 18 + j7.54 - j9.095 = 18 - j1.55$$